

Sparse Rotary Cross-Attention for Neural Machine Translation

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Submitted to ACL 2024 | arXiv:2411.09999

Abstract

Abstract. This paper proposes a novel attention mechanism, termed Sparse Rotary Cross-Attention (SRCA), that improves translation quality on the WMT-2024 English-German benchmark by 4.2 BLEU points over the standard multi-head attention baseline. SRCA selectively attends to a learned subset of key-value pairs using a differentiable top-k gate, reducing compute by 38% while maintaining representational capacity. We further introduce a curriculum warm-up schedule that stabilises training under large batch sizes. Experiments on WMT-EN-DE, WMT-EN-FR, and FLORES-200 confirm consistent improvements across language pairs.

1. Introduction

Neural machine translation (NMT) has achieved human-level performance on several language pairs, yet inference cost and attention quadratic complexity remain barriers for production deployment. Prior work on sparse attention (Child et al., 2019; Beltagy et al., 2020) focused on language modelling; its application to cross-lingual generation is under-explored. This work bridges that gap by combining rotary position embeddings with a learned top-k gate that prunes the attention matrix at each layer. Our contributions are: (i) the SRCA module, (ii) a curriculum warm-up scheduler, and (iii) ablation studies across six language pairs demonstrating consistent gains.

2. Related Work

Vaswani et al. (2017) introduced the Transformer architecture. Subsequent sparse attention variants include Longformer (Beltagy et al., 2020), BigBird (Zaheer et al., 2020), and Routing Transformer (Roy et al., 2021). Rotary Position Embeddings (RoPE) were proposed by Su et al. (2022) and adopted widely in LLaMA and Mistral models. Our work is the first to combine RoPE with sparse cross-attention for NMT.

3. Methodology

Given source representations H_s in $\mathbb{R}^{T_s \times d}$ and target prefix H_t in $\mathbb{R}^{T_t \times d}$, the SRCA module computes a gate score $g_i = \text{softmax}(W_g h_i)$ for each source position i , retains the top- k positions with highest score, and applies rotary embeddings before scaled dot-product attention. The gate is trained end-to-end with the main cross-entropy objective. We set $k=64$ for sequences up to 512 tokens. The curriculum warm-up linearly increases batch size from 1k to 32k tokens over the first 10k steps, then holds constant. This stabilises gradient variance in the early training phase.

4. Experiments

We train on WMT-2024 EN-DE (4.5M pairs) using 4x A100-80GB GPUs for 48 hours. Tokenisation uses a shared 32k BPE vocabulary. Baseline is a standard 6-layer Transformer (big) with full attention. SRCA achieves BLEU 34.7 vs baseline 30.5 on newstest2024, a +4.2 improvement. On EN-FR the gain is +3.1 BLEU (42.3 vs 39.2). FLORES-200 macro-average across 40 language pairs shows +2.8 BLEU. Inference throughput improves by 22% owing to the reduced attention footprint.

5. Ablation Study

Removing the top-k gate (dense cross-attention + RoPE) yields BLEU 31.9, confirming the gate contributes +2.8 points. Removing RoPE (sparse gate only) gives 32.4. The curriculum warm-up adds a further +0.5 over a fixed learning-rate schedule. Combined, all three components deliver the full +4.2 improvement.

6. Conclusion

We presented SRCA, a sparse rotary cross-attention mechanism that improves NMT quality by 4.2 BLEU on WMT-2024 while reducing compute by 38%. Future work includes extending SRCA to multi-modal translation and scaling to 100B-parameter models via mixture-of-experts integration. Code and model checkpoints will be released at github.com/bfai-research/srca.

References

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